# Elyas Saif

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## EDUCATION

**PhD Economics** 2027 (expected)
Hanken School of Economics, Finland

**MSc Economics** 2016
Hanken School of Economics, Finland

**BA Economics** 2013
Wesleyan University, USA

## RESEARCH INTERESTS

Application of AI/ machine learning in Economics, Prediction Policy Problems, and Household Finance.

## JOB MARKET PAPER

**How Predictable is Household Financial Distress?**

I use machine learning (ML) methods to examine the forward-looking predictability of consumer financial distress using nationally representative Finnish household-level data. This predictive study draws parallels with the literature on consumer default predictability but is more aligned with the perspective of the consumer rather than the lender. My study contributes to the literature on consumer financial protection — which focuses on causally identifying measures that mitigate suboptimal borrowing decisions and consumer financial distress — by presenting a predictive dimension that complements this causal focus. With accurate predictions, consumer protection measures can be targeted to households at risk of consumer financial distress. This study provides insights into the overall accuracy of ML in predicting consumer financial distress, revealing significant heterogeneity in predictability between households with and without prior experience of distress. In a budget-constrained financial counselling context, predictability is nearly perfect for households with prior distress but significantly lower for those without. This demonstrates that predictions of consumer financial distress can accurately target support measures for households most likely to experience persistent consumer financial distress.

## WORKING PAPERS

**Interhousehold Private Monetary Transfers: Recipient Financial Distress and Forecast Errors**

**Interhousehold Giving and Unexpected Financial Shocks**

## SEMINAR PRESENTATIONS

Helsinki GSE Labor and Public Economics PhD Seminar 2026
Helsinki GSE Industrial Organization PhD Seminar 2025
Hanken Economics Seminar, Hanasaari Island, Espoo 2024
Helsinki GSE Industrial Organization PhD Seminar 2023
Helsinki GSE Industrial Organization PhD Seminar 2022

## GRANTS

OP Pohjola Research Foundation — €29,000 2026, 2025, 2023, 2022
Hanken School of Economics — €25,000 2024
Yrjö Jahnsson Foundation — €24,000 2021

## REFERENCES

**Ari Hyytinen**
Professor of Economics
Hanken School of Economics
ari.hyytinen@hanken.fi

**Benjamin Maury**
Wahlroos Professor of Finance
Hanken School of Economics
benjamin.maury@hanken.fi

**Topi Miettinen**
Professor of Economics
Hanken School of Economics
topi.miettinen@hanken.fi

## TEACHING

**University Teacher** 2026, 2025, 2024, 2023, 2022

Quantum I: Data-Driven Decision Making

*Lecture topics:*

- Introduction to Python Programming
- Data Analysis with pandas
- Agentic AI Workflows for Data Analysis
- Database Querying with SQL
- Integrating Python, SQL, and BI Tools
- Automating Data Updates with SQL

**Guest Lecturer** 2025, 2024

Analytics Across Business Domains

*Lecture topics:*

- Python Fundamentals for Machine Learning
- Predictive Machine Learning in Economics

## PROFESSIONAL EXPERIENCE PRIOR TO PHD

**Financial Controller** Sept 2017 – Dec 2018

Nissan Nordic Europe, Espoo, Finland

- **Financial Reporting:** Generated financial reports using Excel (VLOOKUP & PivotTable)
- **Report Automation:** Reduced Excel report preparation time by 75% via SQL automation
- **Financial Analysis:** Analyzed transaction data, supported audits, and identified duplicate payments, saving EUR 10K
- **Process Documentation:** Documented data-processing workflows to ensure reproducibility
- **Cross-Functional Collaboration:** Worked with different departments on reporting, customer claims and inquiries, and preparing marketing campaigns

**Paid Research Project (AR/VR in Real Estate Project)** Feb 2017 – May 2017

Aalto University in collaboration with Nokia Corp., Espoo, Finland

- **Stakeholder Engagement & Requirements Gathering:** Interviewed real-estate stakeholders to understand their processes, identify needs, and document requirements for AR/VR solutions
- **Data Analysis:** Analyzed interview data, identifying patterns and insights for Nokia

**Assistant Business Analyst — Summer Internship** Jun 2015 – Aug 2015

Forte Group Inc., Minsk, Belarus

- Researched Belarusian IT companies' core competencies and markets for their services; presented findings to the regional manager.
- Compared wage growth in the Belarusian IT industry with that of other countries.

## SKILLS

**Stata** (**Advanced**): Constructing a longitudinal panel dataset from annual waves of Statistics Finland's Income Distribution Survey (IDS) using unique household identifiers; data cleaning & variable construction (using loops), econometric analysis using fixed effects, DiD, and 2SLS

**Python (Advanced):** NumPy, pandas, scikit-learn, Matplotlib, Seaborn, and imblearn

**SQL (Advanced):** SQL database design and data queries

**Power BI / Power Platform (Advanced):** Power Query, data modelling, Dataverse schema, and Dataverse–Power BI integration

**Excel (Advanced):** VLOOKUP, filters, aggregation functions, PivotTables, and macros

**AI Assisted Development (Advanced):** Rapid prototyping with Agentic AI

**Git/GitHub (Intermediate):** Version control and project documentation

R **programming** (**Intermediate**)

**Languages:** English: full working proficiency; Arabic: native; Russian: native; Italian: conversational; German: conversational; Finnish: basic